

ASSIGNMENT 1

Total Marks 60

Semester: SUMMER 2024

Faculty: RFH

Course No: CSE 250

Section: 11

Softcopy Submission Deadline: 2 PM, 9 July 2024

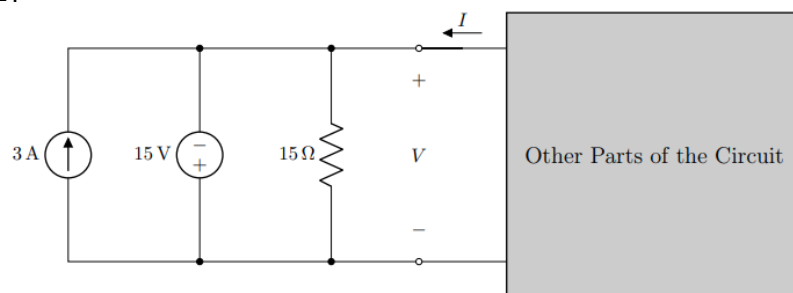
Hardcopy Submission Deadline: 9 July 2024

Softcopy Submission form: <https://forms.gle/oJn2LeaYjY15Gz429>

IV Characteristics

Question 1 [CO2]

2.5+2.5

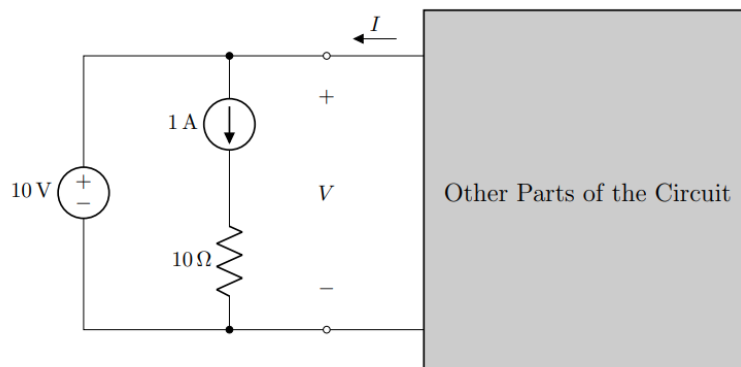


(a) Determine $I - V$ characteristics of the left part of the circuit. i.e. Write an equation that will always hold irrespective of what is inside the greyed-out box. Hint: The equation cannot have any variables other than V and/or I . Everything else should just be a number.

(b) Based on your answer in (a), plot the $I - V$ characteristics of the sub circuit (*No need of graph paper*)

Question 2 [CO2]

2.5+2.5

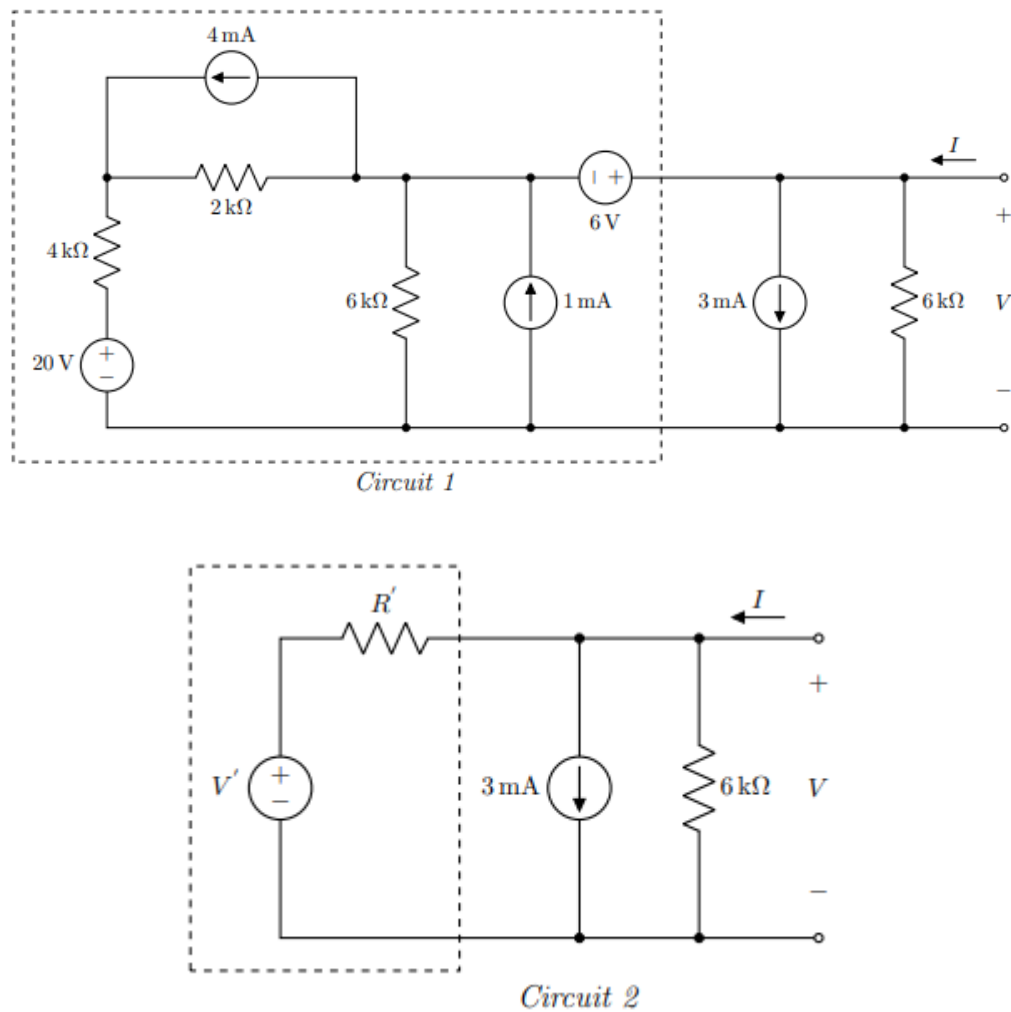


(a) Determine $I - V$ characteristics of the left part of the circuit. i.e. Write an equation that will always hold irrespective of what is inside the greyed-out box. Hint: The equation cannot have any variables other than V and/or I . Everything else should just be a number.

(b) Based on your answer in (a), plot the $I - V$ characteristics of the sub circuit (*No need of graph paper*)

Question 3 [CO2]

5+5

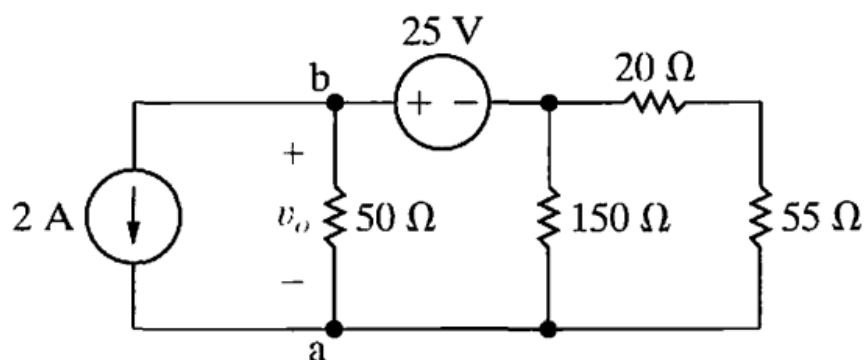


- (a) Apply Source Transformation to reduce the dashed boxed portion of Circuit 1 to a single voltage source in series with a resistor as shown in Circuit 2. What are the values of V' and R' ?
- (b) Derive a Current-Voltage Relationship from Circuit 2. The $I - V$ equation cannot contain any variables other than I and V pointed out in the diagram. Plot the $I - V$ relation in the grid provided above. (No need of graph paper)

Nodal Analysis

Question 4 [CO3]

4+4+2



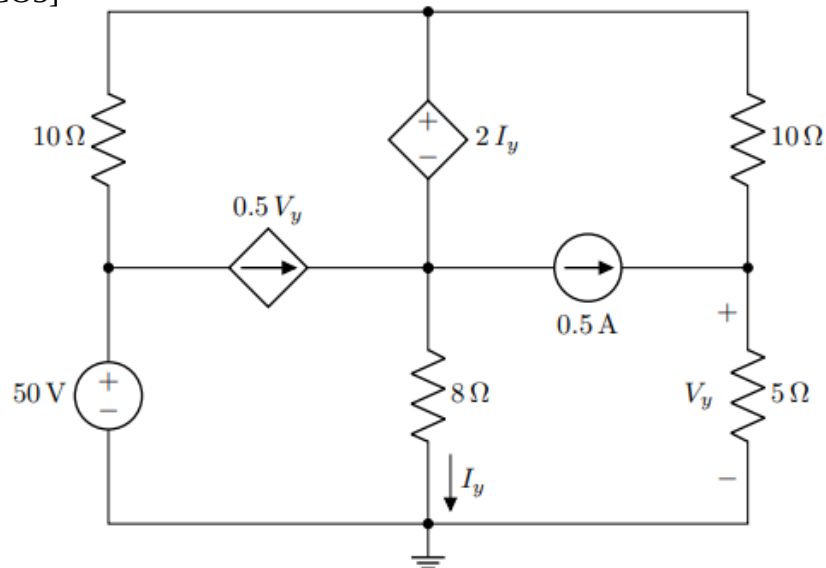
(a) Use the node-voltage method to find v_0 and power delivered by the 2 A current source in the circuit. Use node a as the reference node.

(b) Repeat part (a), but use node b as the reference node.

(c) Compare the choice of reference node in (a) and (b). Which is better, and why?

Question 5 [CO3]

4+2

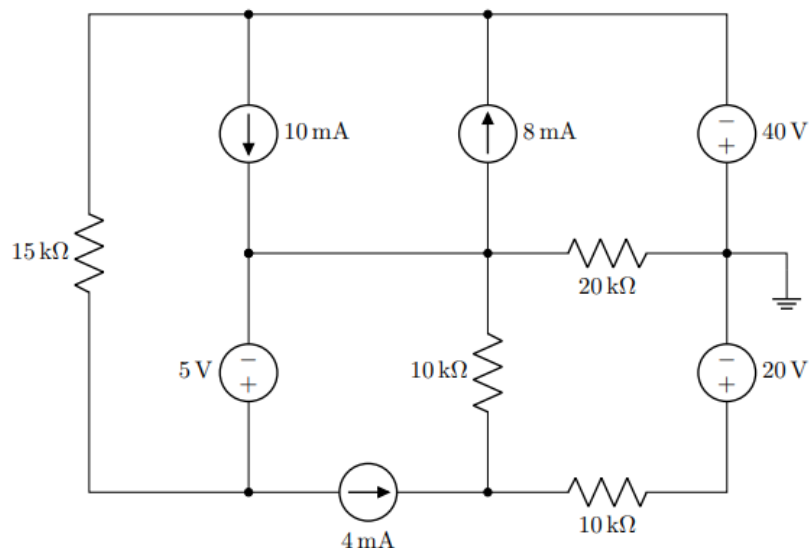


(a) Find all the node voltages in the circuit shown above.

(b) Determine the power of the $2 I_y$ dependent voltage source (with appropriate $\pm$ sign). Also mention, whether the source is supplying or consuming the power.

Question 6 [CO3]

4+2



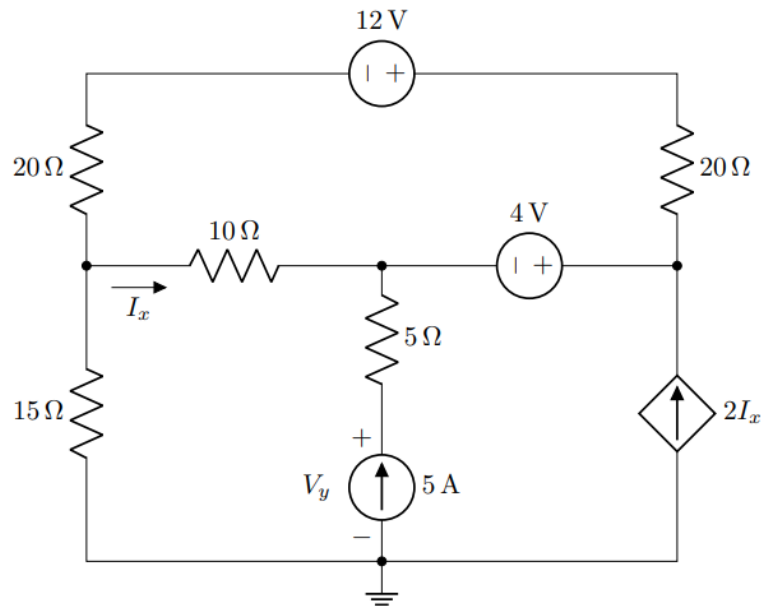
(a) Find all the node voltages in the circuit shown above.

(b) Determine the power associated with the 5 V source in the circuit (with appropriate $\pm$ sign and unit). Also, specify whether the power is being supplied/consumed.

Mesh Analysis

Question 7 [CO3]

4+2+2

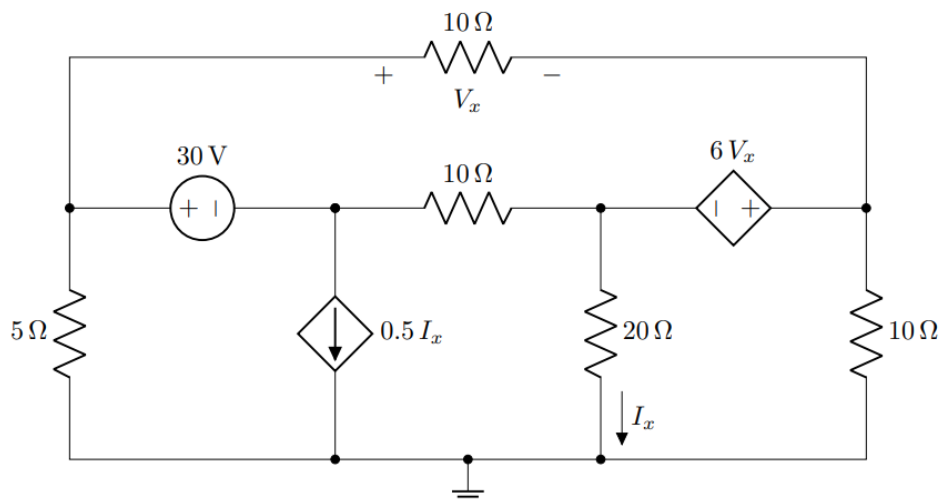


- Find all the node voltages/mesh currents in the circuit.
- Find V_y , the voltage across the 5 A current source.
- How much power is the 5 A current source consuming/supplying to the circuit? Also mention whether the source is supplying or consuming power.

Nodal & Mesh Analysis

Question 8 [CO3]

4+4+2



- Find all the node voltages in the circuit using Nodal Analysis.
- Find all the mesh currents in the circuit using Mesh Analysis.
- Determine the voltage across the $0.5I_x$ dependent current source.